

**International Institute of Information Technology
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Computer Science and Engineering

EcoTwin

**Ecosystem Analog Search and Forecasting Using
Geospatial Foundation Models**

Project Report

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Abstract

EcoTwin is an end-to-end geospatial ecosystem analog discovery platform designed around the question: *which ecosystems elsewhere exhibit similar spatial structure, and what can their historical trajectories tell us about a target ecosystem?* HydroLAKES polygons provide ecosystem boundaries; a buffered, tiled Sentinel-2 acquisition pipeline through Google Earth Engine yields one-kilometre analysis cells, converted into six-band Prithvi embeddings, coverage-pooled to one 768-dimensional lake-year vector and indexed in PostgreSQL/pgvector. A query ecosystem is compared as a *five-year trajectory* rather than a single snapshot, sliding across each candidate’s history to find its best-matching window, and the ranked results are exposed through an interactive map and profile views with historical index trajectories.

The deployed system holds a complete, fully processed cohort — **829 active lakes across 26 Indian states**, ten consecutive years (2016–2025), **8,227 lake-year embeddings**, **19,046** imagery tiles, and **233,002** analysis cells, with no outstanding failures at any pipeline stage. The forecasting layer is analog-based and non-parametric: rather than decoding an embedding back into reflectance, it uses the embedding space only to *rank* analogs, then transfers their measured year-on-year changes onto the target as a rank-weighted directional forecast, traceable to the specific lakes, years, and deltas that produced it. The system is demonstrated on five Ramsar-designated wetlands — Chilika, Wular, Loktak, Tso Morari, and Vembanad — reported in full in Chapter 6.

Keywords: Geospatial Foundation Models, Prithvi, Remote Sensing, Sentinel-2, HydroLAKES, Ecosystem Analogs, Vector Search, pgvector, NDCI, Red-Edge Indices, Ramsar Wetlands, Trajectory Similarity

Contents

Abstract	1
1 Introduction	5
1.1 Background and Problem Statement	5
1.2 Core EcoTwin Hypothesis	5
1.3 Project Evolution and Contributions	6
2 Data Sources and Data Engineering	7
2.1 Overview	7
2.2 Sentinel-2 and Temporal Coverage	8
2.3 Environmental Indices: A Band-Honest Index Set	9
3 Detailed EcoTwin Methodology	11
3.1 End-to-End Pipeline	11
3.2 Data-Fetching Workflow: The Seven Operator-Triggered Stages	11
3.3 Steps 1–2: Lake Boundary Retrieval and Buffering	12
3.4 Step 3: Projected Grid Generation	13
3.5 Steps 4–8: Sentinel-2 Compositing, Export, and Resumability	14
3.6 Step 9: Local Raster Merging	15
3.7 Step 10: Analysis-Cell Generation	15
3.8 Step 11: Model Input Preparation	16
3.9 Step 12: Prithvi Embedding Generation	16
3.10 Step 12a: Coverage-Weighted Lake-Level Pooling	17
3.11 Step 15: Trajectory Similarity Search	17
3.12 Step 16: Temporal Profile Generation	19
3.13 Step 17: Directional Ecological Forecasting	19
3.13.1 Worked Example: Chilika, NDVI-B7, $h = 1$	20
3.13.2 Hindcasting Protocol	20

4	System Architecture and Implementation	22
4.1	Architecture Overview	22
4.2	Technology Stack	23
4.3	Component Communication Workflow	23
4.4	PostgreSQL, PostGIS, and pgvector	24
4.5	Core Database Entities	24
4.6	Backend and Frontend	25
4.7	Asynchronous Processing and Progress Tracking	26
4.8	Server-Side B8 Acquisition	26
4.9	Automated PDF Report Generation	26
4.10	Configuration Management	27
5	Experimental Results and Engineering Evaluation	28
5.1	Deployed Processing Cohort	28
5.2	Pipeline Completion Statistics	29
5.3	Similarity Search Performance	30
5.4	The Score Compression Problem	31
5.5	Temporal Profile Results	31
5.6	Processing Reliability	31
5.7	Engineering Problems Encountered	32
6	Applied Analysis: Five Ramsar Wetlands	33
6.1	Purpose and Site Selection	33
6.2	Chilika Lagoon, Odisha	34
6.2.1	Retrieved Analogs	34
6.2.2	Measured History	34
6.2.3	Forecast	35
6.2.4	Interpretation	35
6.3	Wular Lake, Jammu & Kashmir	35
6.3.1	Retrieved Analogs	35
6.3.2	Measured History — The Clearest Signal in the Study	36
6.3.3	Forecast	36
6.3.4	Interpretation	36
6.4	Loktak Lake, Manipur	37
6.4.1	Retrieved Analogs	37

6.4.2	Measured History	37
6.4.3	Forecast — and an Instructive Failure	38
6.4.4	Interpretation	38
6.5	Tso Morari, Ladakh	38
6.5.1	Retrieved Analogs — The Strongest Validation	38
6.5.2	Measured History	39
6.5.3	Forecast	40
6.5.4	Interpretation	40
6.6	Vembanad, Kerala	40
6.6.1	Retrieved Analogs	41
6.6.2	Measured History	41
6.6.3	Forecast — Degrading Confidence	41
6.6.4	Interpretation	42
6.7	Cross-Site Synthesis	42
7	Discussion	44
7.1	Strengths and Limitations	44
8	Conclusion and Future Work	45
8.1	Conclusion	45
8.2	Future Work	45
8.3	Final Remarks	46

Chapter 1

Introduction

1.1 Background and Problem Statement

Earth observation satellites provide repeated measurements of ecosystems at spatial scales that cannot be obtained economically through field observations alone, and Sentinel-2’s multispectral optical imagery is well suited to vegetation, water, land-cover, and surface-condition analysis. Rather than beginning with a specific question such as vegetation change or water extent, EcoTwin treats ecosystem comparison as a retrieval problem: satellite observations are converted into learned feature vectors and indexed, so a lake can be compared against a large reference collection without repeatedly processing raw imagery, asking whether another ecosystem looks similar in its spatial-spectral representation.

Several engineering constraints shape how this is actually solved. Lake boundaries are irregular polygons rather than fixed-size image squares, and large satellite scenes are expensive to download and store, so large ecosystems may require multiple image tiles rather than a single export. Historical Sentinel-2 collections do not expose identical quality bands across all years, and cloud or invalid-pixel contamination must be controlled for regardless of collection; every resulting embedding must remain associated with its source geometry and lake/year metadata rather than existing as an anonymous vector, and online similarity search must be substantially cheaper than recomputing embeddings from scratch.

1.2 Core EcoTwin Hypothesis

EcoTwin is based on the operational hypothesis:

Ecosystems elsewhere that are structurally similar in geospatial feature space can provide useful analogs for understanding the spatial and historical behaviour of a target ecosystem.

The hypothesis separates two tasks. The first is **analog discovery**: finding ecosystems that resemble the target in the learned spatial-spectral representation. The second is

trajectory analysis: comparing the historical environmental indicators of the target against those of its discovered analogs.

1.3 Project Evolution and Contributions

The implementation evolved from an initial design of standardized 5×5 km ecosystem blocks toward a lake-centred buffered imagery stage, local raster merging, and 1×1 km analysis cells: the larger imagery tiles are an I/O and Earth Engine management mechanism, while the smaller cells are the units from which embeddings are generated, which lets large lakes be processed without forcing the foundation model to operate directly on irregular polygons.

Two further pivots shaped the final system. First, **snapshot similarity gave way to trajectory similarity:** comparing single query and candidate embeddings proved too weak a notion of ecosystem resemblance, since two lakes can coincide in one year and diverge over a decade, so the implemented system instead compares a *five-year trajectory* of the query against every five-year sliding window in each candidate’s history, scoring by the best-matching window and retaining its position in time (Section 3.11). Second, an **embedding decoder was designed and then abandoned:** the original design intended to decode a predicted future embedding back into spectral indices, which would have required training an inversion of Prithvi’s 768-dimensional space — not calibrated, and exactly the kind of opacity the analog architecture exists to avoid. The implemented layer instead uses embeddings *only* to rank analogs, and computes predicted quantities from the analogs’ measured band statistics already in the database (Section 3.13): a transfer of observed change, not a reconstruction of an imagined image.

The project’s contribution is the resulting end-to-end lake-to-embedding-to-analog pipeline, executed to completion over an 829-lake, ten-year cohort with no residual failures: a satellite-image acquisition strategy built on Google Earth Engine and local raster processing with polygon-aware spatial subdivision and coverage filtering; Prithvi-based six-band representations with coverage-weighted pooling from cell to lake level, persisted as 768-dimensional vectors and retrieved through trajectory-based analog search; the decoder-free forecasting method above; a band-honest ecological index set computed only from channels the imagery actually contains; and a FastAPI/React-Leaflet application with automated PDF reporting, evaluated on five Ramsar wetlands of international importance.

Chapter 2

Data Sources and Data Engineering

2.1 Overview

EcoTwin combines static geographic information (HydroLAKES polygons and meta-data) with repeated satellite observations (Sentinel-2), so that the static layer defines *where* an ecosystem is while the satellite layer defines *what it looks like through time*. The database representation retains the identifying, geographic, and descriptive fields it needs downstream — lake identifier and name, country, polygon geometry, area, elevation, pour-point coordinates, lake type, average depth, total volume, and watershed area — summarised with their pipeline role in Table 2.1. The complete HydroLAKES-derived inventory is much larger than the processing cohort: the pipeline was narrowed to a selected subset of lakes so that multi-year imagery and embeddings could be generated within available storage, Earth Engine, and compute budgets.

Table 2.1: HydroLAKES-derived information used by EcoTwin.

Field	Type	Role in EcoTwin
Lake ID	Integer	Primary ecosystem identifier
Lake name	Text	Human-readable identification
Country	Text	Geographic filtering
Lake area	Numeric	Size and metadata analysis
Elevation	Numeric	Environmental context
Pour latitude/longitude	Numeric	Geographic reference
Geometry	PostGIS geometry	Spatial processing and mapping
Lake type	Categorical	Ecosystem metadata
Average depth	Numeric	Descriptive context
Total volume	Numeric	Descriptive context
Watershed area	Numeric	Catchment context

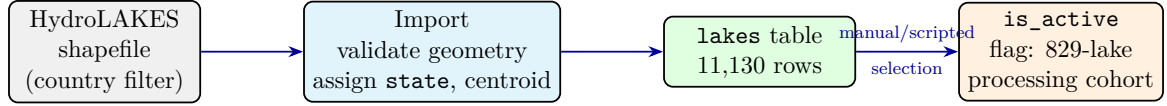


Figure 2.1: Stage 1 (Sync Lakes): the HydroLAKES shapefile is imported once per country into the full `lakes` inventory (11,130 rows), and `is_active` marks the 829-lake processing cohort that the remaining six stages operate on.

2.2 Sentinel-2 and Temporal Coverage

Sentinel-2 multispectral imagery, filtered, composited, and exported through Google Earth Engine, is the primary remote-sensing input. The principal collection is `COPERNICUS/S2_SR_HARMONIZED`, with a fallback to `COPERNICUS/S2` where the surface-reflectance harmonized collection provides no observations — particularly important for early Sentinel-2 years.

Ingestion was extended from an initially planned 2018–2025 window back to 2016, giving ten consecutive years. Table 2.2 shows the result: from 2018 onward every one of the 829 active lakes has a complete, embedded, feature-extracted record, while 2016 and 2017 fall slightly short because Sentinel-2B did not launch until March 2017, leaving a single-satellite, longer-revisit period in which some lakes lacked enough cloud-free acquisitions for a usable annual composite.

Table 2.2: Achieved temporal coverage across the 829-lake cohort. Every retained lake-year carries both an embedding and a completed feature record; there are no partial rows.

Year	Lake-years	Embeddings	Features
2025	829	829	829
2024	829	829	829
2023	829	829	829
2022	829	829	829
2021	829	829	829
2020	829	829	829
2019	829	829	829
2018	829	829	829
2017	808	808	808
2016	787	787	787
Total	8,227	8,227	8,227

The 42 absent lake-years are recorded as such rather than interpolated, since the similarity layer requires five consecutive years to form a comparison window: a lake missing 2016 simply contributes fewer candidate windows rather than a fabricated one.

The embedding input uses six bands, B2–B7 (blue, green, red, narrow near-infrared, and two short-wave infrared channels), selected consistently from the exported GeoTIFFs before model inference.

2.3 Environmental Indices: A Band-Honest Index Set

The merged lake imagery contains only B2–B7 — not B8 (broad NIR), B11, or B12 (SWIR) — so the textbook forms of NDVI, NDWI, and NBR, which all require B8 or SWIR, cannot be computed. Computing them from substituted bands while still calling them NDVI, NDWI, or NBR would misreport what was measured, so EcoTwin instead uses five indices chosen to be fully supported by the bands actually present and named for what they really are, computed from the per-band *mean* reflectance of each lake-year over pixels inside the lake polygon: NDCI ($\frac{B5-B4}{B5+B4}$), the primary eutrophication/algal-bloom indicator, since B5 (705 nm, red edge) is where chlorophyll-a reflectance rises sharply while B4 (665 nm) remains absorbed; NDVI-B7 ($\frac{B7-B4}{B7+B4}$), substituting the absent B8 with the narrow-NIR red-edge channel B7 (783 nm) for a vegetation-vigor measure that tracks floating and emergent vegetation — macrophytes, water hyacinth, and the *phumdi* mats of some Indian wetlands — with the substitution recorded in the name rather than concealed; NDWI ($\frac{B3-B7}{B3+B7}$), the same substitution applied to the water index, which falls as vegetation raises B7 and so acts as NDVI-B7’s natural mirror signal; Turbidity ($\frac{B4-B3}{B4+B3}$), which rises with suspended sediment; and Red-Edge Slope ($\frac{B7-B5}{78\text{nm}}$), the gradient across the red-edge region, sensitive to pigment concentration and canopy density independently of the normalized-difference family.

Table 2.3: Implemented ecological index set and ecological meaning.

Index	Formula	Ecological interpretation
NDCI	$(B5 - B4)/(B5 + B4)$	Chlorophyll-a; eutrophication and algal bloom intensity
NDVI-B7	$(B7 - B4)/(B7 + B4)$	Vegetation vigor; macrophyte and floating-weed encroachment
NDWI	$(B3 - B7)/(B3 + B7)$	Open-water extent; falls as vegetation colonises the surface
Turbidity	$(B4 - B3)/(B4 + B3)$	Suspended sediment load and water clarity
Red-edge slope	$(B7 - B5)/78\text{ nm}$	Pigment concentration and canopy density trend

All five are computed in `app/services/index_calculator.py`, which returns `None` for any index whose input bands are missing rather than substituting a default — so a gap in the data propagates as a gap, never as a spurious zero.

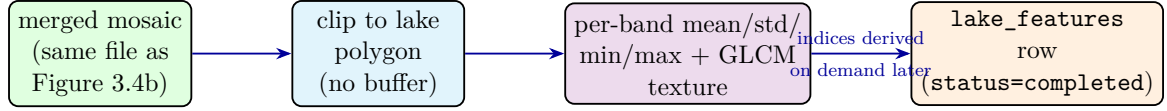


Figure 2.2: Stage 6 (Generate Lake Features): unlike Prithvi inference, this stage clips the merged mosaic to the bare lake polygon (no buffer, no cell grid) and computes per-band statistics and GLCM texture directly; only the band means are persisted, and each index of Table 2.3 is recomputed on demand (Section 3.13).

B8 is deliberately excluded from the Prithvi input, whose band count and normalization constants are fixed by the model’s training configuration, but is useful as an independent descriptor, so it is acquired separately (Section 4.8) and stored alongside the other band statistics for every lake-year.

A valid ecosystem record requires a valid lake polygon and year, a successful imagery query and usable composite, valid exported raster data, valid spatial cells after coverage filtering, and a successfully generated embedding for each retained cell; the pipeline distinguishes successful, failed, skipped, and already-completed work so that long-running jobs can be resumed.

Chapter 3

Detailed EcoTwin Methodology

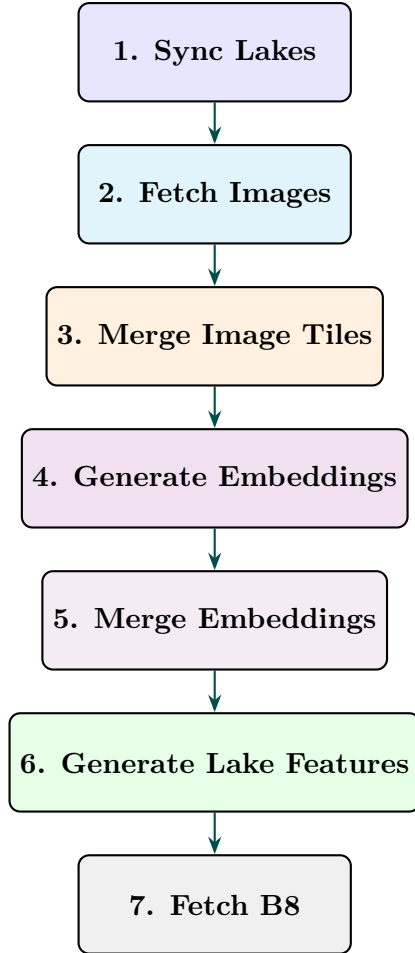
3.1 End-to-End Pipeline

The implemented workflow is:

$$\begin{aligned} & \textit{Lake Inventory} \rightarrow \textit{Boundary Validation} \\ & \quad \rightarrow \textit{Buffering} \rightarrow \textit{Imagery Tiling} \\ & \quad \rightarrow \textit{Sentinel-2 Composite} \rightarrow \textit{GeoTIFF} \\ & \quad \rightarrow \textit{Local Mosaic} \rightarrow \textit{1 km Cells} \\ & \quad \rightarrow \textit{Coverage Filter} \rightarrow \textit{Prithvi} \\ & \quad \rightarrow \textit{768D Embeddings} \rightarrow \textit{pgvector} \\ & \quad \rightarrow \textit{Cosine Search} \rightarrow \textit{Analog} \\ & \quad \rightarrow \textit{Temporal Profiles} \rightarrow \textit{Forecast} \end{aligned} \tag{3.1}$$

3.2 Data-Fetching Workflow: The Seven Operator-Triggered Stages

The seventeen-step scientific pipeline above is what EcoTwin *computes*; Figure 3.1 shows how an operator actually *runs* it, as the seven sequential stages exposed by the control strip of Figure 4.1. Each stage is a separate API call, a separate FastAPI `BackgroundTasks` job, and a separate database write — so a failure at stage 4 leaves stages 1–3 fully intact and does not require re-fetching imagery already on disk.



1. Sync Lakes — import HydroLAKES into the `lakes` table (Figure 2.1).

2. Fetch Images — buffer, tile, and export Sentinel-2 GeoTIFFs per lake-year (Steps 1–8 below).

3. Merge Image Tiles — mosaic a lake-year’s tiles into one raster (Step 9).

4. Generate Embeddings — split the mosaic into cells and run Prithvi inference (Steps 10–12).

5. Merge Embeddings — pool cell embeddings into one lake-year vector (Step 12a).

6. Generate Lake Features — compute per-band statistics and ecological indices (Step 16, Figure 2.2).

7. Fetch B8 — server-side `reduceRegions()` enriches `lake_features` with the auxiliary NIR band (Figure 4.3).

The remainder of this chapter walks through these seven stages at step-by-step granularity, numbered independently of the stage list above.

Figure 3.1: The seven-stage data-fetching workflow, in the order an operator triggers them via the control strip (Figure 4.1). Each stage reads the previous stage’s database output as its own input, never the raw upstream source directly — stage 3 reads `lake_tiles`, not Earth Engine again; stage 6 reads the merged mosaic already on disk, not a fresh GEE export. Stage 7 (B8) is independent of stages 4–5 and can run in parallel with them, since it enriches existing `lake_features` rows rather than depending on embeddings.

Every stage is **idempotent at the row level**: each writes a status column (`pending/processing/completed/failed`) before doing any work, and a re-run skips rows already `completed` rather than reprocessing them. Stage 7 (B8) has no code dependency on stages 4–5 despite the drawn order, since B8 statistics are written directly onto an existing `lake_features` row, so it can be fetched before, after, or concurrently with embedding generation.

3.3 Steps 1–2: Lake Boundary Retrieval and Buffering

The pipeline reads the target lake polygon from PostGIS, keeping the original geometry in a spatial database to avoid repeated conversion between shapefiles and application-

specific formats, checks it for validity, and transforms it into a projected CRS. The polygon is then expanded using a configurable buffer (1200 m during development), since the ecosystem representation is not restricted to the water polygon itself. The surrounding land, shoreline, vegetation, and local landscape context contribute to the spatial-spectral representation.

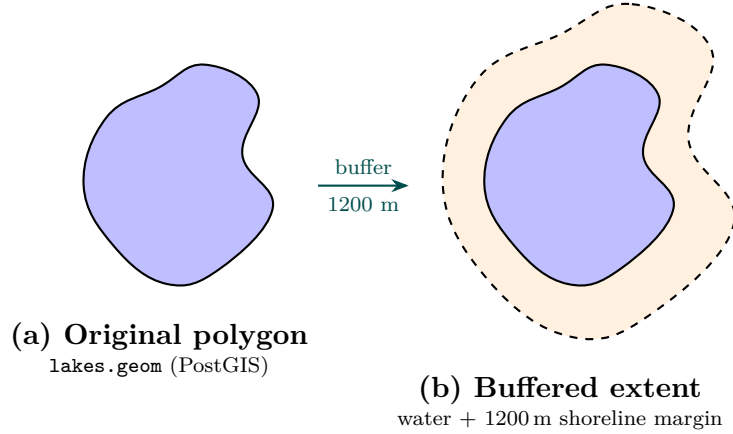


Figure 3.2: Step 2 (Buffering): the original polygon (a) is expanded by a fixed 1200 m buffer (b); everything between the two boundaries is retained as part of the ecosystem representation, not discarded as “outside the lake.”

3.4 Step 3: Projected Grid Generation

The buffered polygon is projected into a metric CRS. A regular imagery grid is then generated around the buffered geometry.

The imagery grid is deliberately different from the final analysis grid. During the latest extraction implementation, an approximately 8500 m tile size was used to balance Earth Engine request size, memory use, and local processing cost.

A lake may therefore produce one, two, or several imagery tiles depending on its size and shape.

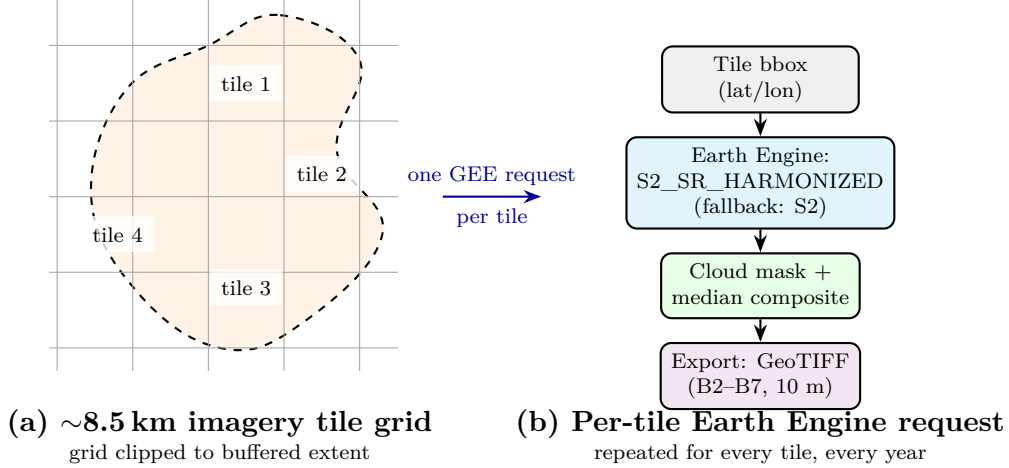


Figure 3.3: Step 3 (Imagery Tiling): the buffered extent (a) is clipped against a ~ 8.5 km grid (here four tiles); each tile drives one Earth Engine request (b) — the “Fetch Images” data-fetching stage of Figure 3.1.

3.5 Steps 4–8: Sentinel-2 Compositing, Export, and Resumability

For each lake-year combination, the pipeline first queries the preferred surface-reflectance collection; if no usable observation exists, it falls back to the Level-1C collection (Table 3.1). This decision is made before the export request and is recorded in the processing metadata.

Table 3.1: Sentinel-2 collection strategy.

Priority	Collection	Purpose
1	COPERNICUS/S2_SR_HARMONIZED	Preferred surface reflectance
2	COPERNICUS/S2	Fallback for early or unavailable surface-reflectance years

Scenes are filtered before compositing: the deployed configuration excludes those whose reported cloud cover exceeds 20% (`GEE_CLOUD_COVER_MAX`) and masks individual unusable pixels using cloud-probability and class-based information where available — collection-aware, since early Sentinel-2 products expose different quality bands (`QA60`, `MSK_CLASS1_OPAQUE`, `MSK_CLASS1_CIRRUS`) than later ones. The filtered observations are then combined into an annual per-pixel, per-band median composite,

$$I_{\text{annual}}(p, b) = \text{median}(I_1(p, b), I_2(p, b), \dots, I_n(p, b)) \quad (3.2)$$

clipped to the requested extent, and exported as local GeoTIFF at 10 m resolution in floating-point form, so the same composite never needs recomputing for a later analysis cell. Each tile carries status metadata (lake ID, year, tile index, bounding box,

local path), and the pipeline checks this before starting a new Earth Engine request for the same tile — concrete, tile-level idempotency (Section 3.2): a batch interrupted partway through a lake resumes from its first incomplete tile.

3.6 Step 9: Local Raster Merging

After all tiles belonging to a lake-year are available, rasterio merges them into a single raster mosaic — reading, reprojecting, and writing with compression settings that reduce storage while preserving the numerical data model inference needs. The merged image becomes the stable source from which smaller analysis cells are extracted.

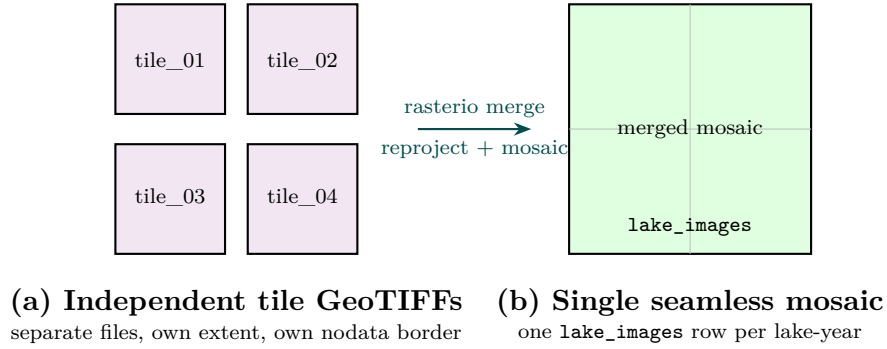


Figure 3.4: Step 9 (Local Raster Merging): independent tile GeoTIFFs (a) are mosaicked by `rasterio.merge` into one seamless lake-year raster (b), recorded in `lake_images` — the “Merge Image Tiles” stage of Figure 3.1.

3.7 Step 10: Analysis-Cell Generation

The merged lake image is intersected with a regular 1×1 km analysis grid, and for each cell the lake coverage is computed,

$$Coverage(c) = \frac{Area(c \cap Lake)}{Area(c)} \times 100 \quad (3.3)$$

with only cells above a 15% threshold retained, which prevents cells that only touch a small corner of a lake from being treated as representative ecosystem regions.

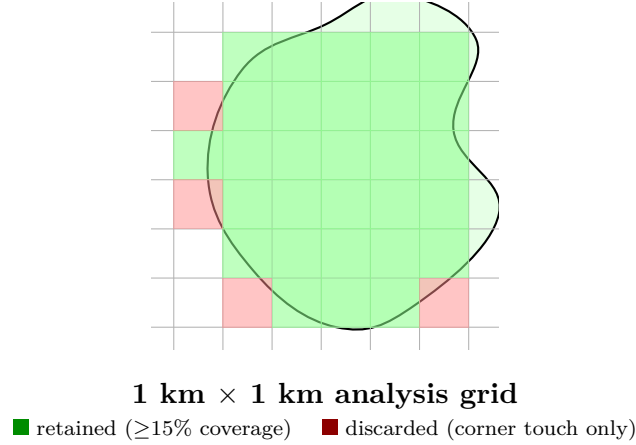


Figure 3.5: Step 10 (Analysis-Cell Generation): the merged mosaic is intersected with a 1 km grid; cells with substantial coverage (green) are retained as embedding units, corner-touch cells below 15% (red) are discarded.

3.8 Step 11: Model Input Preparation

Each retained analysis cell is read from the local merged GeoTIFF and the six selected bands are converted into the model’s expected tensor layout, numerical range, and spatial dimensions. Separating geographic geometry, raster extraction, model preprocessing, and model inference into distinct stages makes it possible to change the model without re-downloading the underlying satellite data.

3.9 Step 12: Prithvi Embedding Generation

Prithvi is used as the geospatial foundation model: each retained analysis cell, as a six-band B2-B7 image X , is transformed into a fixed-length feature vector $f(X) \rightarrow \mathbf{z} \in \mathbb{R}^{768}$, batched eight at a time (EMBEDDING_GPU_BATCH_SIZE) on CPU inference. Every output vector is L2-normalised, so that for unit-norm vectors the cosine similarity reduces exactly to the inner product and the entire similarity search can be executed as matrix multiplication.

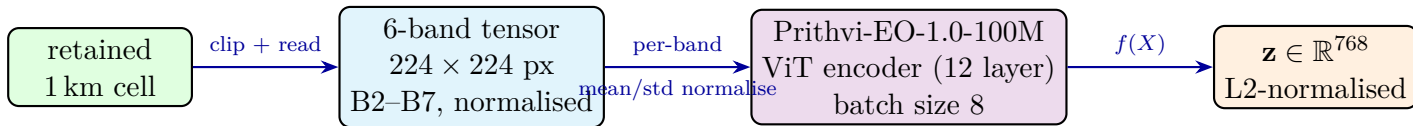


Figure 3.6: Steps 11–12: each retained cell is clipped and read as a 224×224 -pixel, six-band tensor, normalised using Prithvi’s own per-band mean/std (PRITHVI_BAND_MEAN/_STD), then batched into the Prithvi-EO-1.0-100M encoder — the only stage where the foundation model is invoked — producing a 768-d, L2-normalised vector.

3.10 Step 12a: Coverage-Weighted Lake-Level Pooling

Cell-level embeddings preserve spatial detail, but retrieval and forecasting operate on lakes, so for a lake-year with cells $c_1, \dots, c_N$ having embeddings $\mathbf{e}_i$ and lake-coverage percentages w_i , the lake embedding is the coverage-weighted mean, re-normalised:

$$\mathbf{z}_{lake} = \frac{\sum_{i=1}^N w_i \mathbf{e}_i}{\left\| \sum_{i=1}^N w_i \mathbf{e}_i \right\|_2} \quad (3.4)$$

so that a cell almost entirely filled by the lake contributes more than one that merely clips its margin; cells with zero coverage, zero vectors, or the wrong dimensionality are excluded before pooling. Across the cohort this reduces **233,002** cell embeddings to **8,227** lake-year vectors, an average of **28.3** cells per lake-year, and the cell-level vectors are retained as the underlying evidence for any result the pooled vector produces.

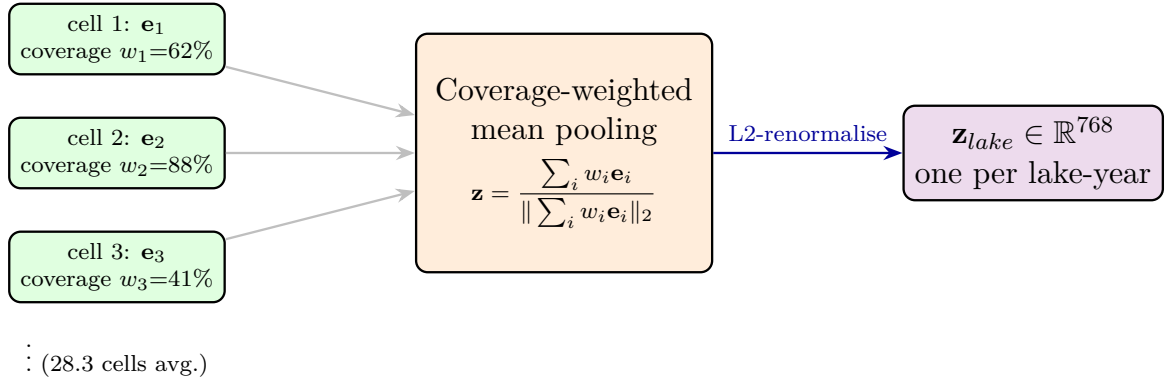


Figure 3.7: Step 12a (Coverage-Weighted Pooling) — the “Merge Embeddings” stage of Figure 3.1: each cell embedding contributes in proportion to its coverage w_i (e.g. 88% outweighs 41%), and the weighted sum is L2-renormalised for comparability in regions.

Before insertion into the database, every embedding is checked for expected dimensionality, missing or null values, numerical validity, and correct lake, year, and spatial-cell association; a failed inference is not inserted as a valid vector. The final embedding is then stored in PostgreSQL using pgvector under the conceptual relationship Lake $\rightarrow$ Year $\rightarrow$ Spatial Cell $\rightarrow$ Embedding, which preserves the provenance required to explain why an ecosystem was retrieved as an analog.

3.11 Step 15: Trajectory Similarity Search

Comparing one query embedding against one candidate embedding asks only *which lake looked like this lake, in this year?* — too weak a question for forecasting, since two lakes can coincide in a single year and diverge over the following decade. EcoTwin instead asks which lake has been **moving through** the representation space the way

this lake has been moving: a shared five-year path is a far better basis for projecting the next three.

For a query lake with embeddings ordered by year, EcoTwin forms the trajectory ending at anchor year t_0 with window size $W = 5$,

$$Q = (\mathbf{q}_{t_0-4}, \mathbf{q}_{t_0-3}, \mathbf{q}_{t_0-2}, \mathbf{q}_{t_0-1}, \mathbf{q}_{t_0}) \quad (3.5)$$

and each candidate lake with at least W years of history is expanded into every contiguous five-year window (ten years of data yields six windows),

$$C^{(j)} = (\mathbf{c}_{y_j}, \dots, \mathbf{c}_{y_j+4}), \quad j = 1, \dots, n - W + 1 \quad (3.6)$$

which need not align in calendar time with the query: a lake that underwent in 2017–2021 the transition another lake is undergoing in 2021–2025 is precisely the analog sought, since it has already lived through the target’s near future. Each window is scored as the mean of the year-by-year cosine similarities between corresponding positions,

$$S(Q, C^{(j)}) = \frac{1}{W} \sum_{i=1}^W \frac{\mathbf{q}_i^\top \mathbf{c}_i^{(j)}}{\|\mathbf{q}_i\|_2 \|\mathbf{c}_i^{(j)}\|_2} \quad (3.7)$$

which (all embeddings being unit-norm) reduces to a mean of inner products, and each candidate lake is represented by its single best window,

$$S_{lake}^* = \max_j S(Q, C^{(j)}), \quad j^* = \arg \max_j S(Q, C^{(j)}) \quad (3.8)$$

whose end year $y_{j^*} + 4$ is the pivot on which the forecasting layer turns: the analog’s observed behaviour *after* that year is the evidence used to project the target’s future.

Because every vector is unit-norm and drawn from one model, raw cosine scores cluster tightly just below 1.0 (measured top analogs span roughly 0.986–0.999 in Chapter 6), which would make every analog read as “99% similar” if displayed directly. EcoTwin therefore linearly rescales $[\phi, 1.0] \rightarrow [0, 1]$ with floor $\phi = 0.5$,

$$S_{display} = \text{clip}_{[0,1]} \left(\frac{S^* - \phi}{1 - \phi} \right) \quad (3.9)$$

a monotone transform that changes no ranking; raw scores are reported alongside the rescaled ones in Chapter 6 so the distinction stays visible. Two further metrics are implemented and selectable per query (Euclidean distance mapped by $1/(1+d)$, and inner-product mapped by $(\langle \mathbf{a}, \mathbf{b} \rangle + 1)/2$), though cosine is the default used throughout this report.

The current implementation compares in Python across all candidate windows: with L lakes, n years, and dimension d , the cost is $O(L \cdot (n-W+1) \cdot W \cdot d)$, roughly $829 \times 6 \times 5 \times 768 \approx 1.9 \times 10^7$ multiply-accumulates per query — measured at **141–172 ms** in Chapter 6, comfortably interactive but linear in lake count, which motivates the pgvector ANN migration under future work.

3.12 Step 16: Temporal Profile Generation

For the query lake and its retrieved analogs, the stored band statistics are assembled into per-year index series using the formulas of Table 2.3, computed on demand from the persisted band means (the single source of truth for measured reflectance) rather than stored pre-computed, so a change to an index definition takes effect everywhere at once without a database migration.

3.13 Step 17: Directional Ecological Forecasting

The forecasting layer transfers *measured change* from the analogs onto the target, presented here in implementation order. The most recent completed feature row for the target lake fixes the anchor year t_0 and anchor values $v_k(t_0)$ for each index k ; forecasts are produced for $t_0 + 1, t_0 + 2, t_0 + 3$ (`_FORECAST_HORIZON`). The trajectory search of Section 3.11 then runs with an enlarged candidate pool, walked in rank order: a candidate is admitted as a *twin* only if it is a different lake than the target, has a completed feature row at its matched window end year, and has at least one usable year after that, so the five twins that emerge are the five best *usable* analogs rather than merely the five highest-scoring rows.

For twin i with matched window ending at year m_i , the change transferred at offset h is $\delta_{i,k,h} = v_{i,k}(m_i + h) - v_{i,k}(m_i)$ — not the analog’s absolute index value, which reflects its own climate, geology, and catchment, but the *change* it underwent, comparable across lakes in a way absolute values are not. Twins receive fixed weights by rank rather than weights derived from their similarity scores,

$$\omega_1 = 0.28, \quad \omega_2 = 0.24, \quad \omega_3 = 0.20, \quad \omega_4 = 0.16, \quad \omega_5 = 0.12 \quad (3.10)$$

precisely because raw similarity scores compress into a narrow band near 1.0 (Section 3.11), where score-proportional weights would be nearly uniform and discard the ranking the search worked to establish. Twins do not all have data at every offset, so with $\mathcal{A}_{k,h}$ the set of twins with a usable delta for index k at offset h , weights are renormalised over exactly that set,

$$\tilde{\omega}_{i,k,h} = \frac{\omega_i}{\sum_{j \in \mathcal{A}_{k,h}} \omega_j}, \quad i \in \mathcal{A}_{k,h} \quad (3.11)$$

so the denominator is always 1.0 and a year with only two twins does not silently shrink toward “stable”; the twin count is reported separately as confidence. The score for index k at offset h is

$$\Sigma_{k,h} = \sum_{i \in \mathcal{A}_{k,h}} \tilde{\omega}_{i,k,h} \delta_{i,k,h} \quad (3.12)$$

classified against threshold $\tau = 0.005$:

$$D_{k,h} = \begin{cases} \text{uncertain (?)} & |\mathcal{A}_{k,h}| < 2 \\ \text{up } (\uparrow) & \Sigma_{k,h} > +\tau \\ \text{down } (\downarrow) & \Sigma_{k,h} < -\tau \\ \text{stable } (\rightarrow) & \text{otherwise} \end{cases} \quad (3.13)$$

with *uncertain* checked first and kept distinct from *stable*: *stable* is a positive finding that the twins agree little changes, while *uncertain* admits too few twins had data to support any claim, and Chapter 6 contains real cases where this distinction carries the entire meaning of a result. An indicative magnitude $\hat{v}_k(t_0 + h) = v_k(t_0) + \Sigma_{k,h}$ accompanies each direction, reported as an expected value rather than a calibrated point prediction. Every forecast returns with a complete audit report — anchor values, weight scheme, classification rule, and each twin’s identity, matched window, and weighted contribution — so any published direction can be recomputed by hand from the reported numbers.

Unlike a parametric time-series model, whose prediction runs through learned parameters difficult to connect to a particular ecosystem, EcoTwin provides an explicit, inspectable chain: Target $\rightarrow$ Retrieved Twin $\rightarrow$ Similarity $\rightarrow$ Historical Trajectory $\rightarrow$ Projected Trajectory. There is no learned parameter anywhere in the method above; every quantity is either a measured reflectance-derived index or a fixed, disclosed constant ($\omega_{1..5}$ and τ), and an interval-based uncertainty measure (the spread across twin predictions) was considered and rejected in favour of the simpler twin count $|\mathcal{A}_{k,h}|$, since a naive dispersion measure over a small, unevenly-weighted set — as when Vembanad’s 2027 red-edge score was dominated by one twin with an extreme delta (Chapter 6) — is not a stable confidence signal, while the count drops to zero exactly when a forecast should not be trusted and degrades gracefully otherwise.

3.13.1 Worked Example: Chilika, NDVI-B7, $h = 1$

Chapter 6 reported Chilika’s 2026 NDVI-B7 forecast as $\uparrow$ with score $+0.0244$ from five twins. Reconstructing it: all five twins have a usable year at $h = 1$, so no renormalization is needed beyond the base weights $(0.28, 0.24, 0.20, 0.16, 0.12)$, which already sum to 1. Each twin’s delta is its measured NDVI-B7 one year after its own matched-window end, minus its value at that end year — five independent measurements from five different lakes and calendar years, combined only through their rank weight. The result, $+0.0244$, exceeds $\tau = 0.005$, giving $\uparrow$, and every intermediate quantity is recoverable from the audit report returned alongside the forecast.

3.13.2 Hindcasting Protocol

A rigorous accuracy evaluation requires hindcasting: hide each lake’s final h_{max} years, run the identical retrieval and forecast pipeline using only years before the cutoff, and score the forecast against the held-out observations. For cutoff year $t_c \in \{2019, \dots, 2022\}$, both the query trajectory and every candidate’s embedding

history are restricted to years $\leq t_c$; the method then runs exactly as deployed, producing $\hat{v}_k(t_c + h)$ for $h = 1, 2, 3$, compared against the actual recorded $v_k(t_c + h)$ already present in `lake_features`. The protocol reports, per index and per horizon (never pooled, since scales differ by an order of magnitude or more), both a magnitude metric and a *directional hit rate*:

$$MAE_{k,h} = \frac{1}{n} \sum_{i=1}^n |v_{k,i}(t_c + h) - \hat{v}_{k,i}(t_c + h)| \quad (3.14)$$

$$HitRate_{k,h} = \frac{1}{n} \sum_{i=1}^n \mathbb{I}[\text{sign}(\Sigma) = \text{sign}(\Delta_{actual})] \quad (3.15)$$

Running this at $t_c = 2022$ requires three held-out years (2023–2025) plus five years of prior history — exactly the ten years this project has, so a single cutoff gives only one evaluation point per lake: too coarse a sample for a defensible aggregate accuracy figure. Multiple cutoff years are not yet available because the dataset spans only one decade, so this protocol is documented here as the immediate next step (Section 8.2) rather than executed prematurely on insufficient data.

Chapter 4

System Architecture and Implementation

4.1 Architecture Overview

EcoTwin is implemented as a multi-component system. The offline pipeline performs the computationally expensive work, while the online application retrieves precomputed data.

Figure 4.1 shows the pipeline control strip from the deployed application. Each button triggers one of the seven offline stages of Figure 3.1 (Section 3.2), in the same execution order. Exposing each stage as an individually triggerable operation, rather than one opaque “run everything” action, is what makes the pipeline resumable (Step 8, Chapter 3): a failure in one stage does not require re-running the stages before it.

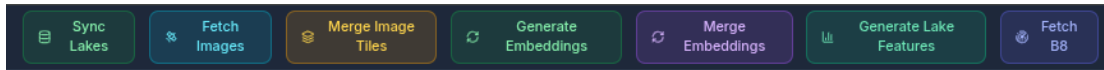


Figure 4.1: The deployed pipeline control strip. Each stage (lake sync, imagery fetch, tile merge, embedding generation, embedding merge, feature generation, B8 fetch) is independently triggerable and independently resumable.

4.2 Technology Stack

Table 4.1: EcoTwin implementation stack.

Layer	Technology	Responsibility
Frontend	React 18, Vite, TypeScript	Interactive application
Mapping	Leaflet / React-Leaflet	Spatial visualization
Charts	Recharts	Index trajectory plots
Client state	Zustand, TanStack Query	Local state and server cache
Backend	FastAPI, Python 3.12	REST services and orchestration
ORM / migrations	SQLAlchemy 2 (async), Alembic	Schema and access
Database	PostgreSQL 16	Relational persistence
Spatial database	PostGIS 3.4	Polygon and spatial queries
Vector search	pgvector	Embedding storage and similarity search
Remote sensing	Google Earth Engine	Satellite filtering/compositing
Raster processing	Rasterio, NumPy, Shapely	Local GeoTIFF processing
ML inference	Prithvi-EO-1.0-100M, PyTorch	Geospatial embeddings
Background work	FastAPI BackgroundTasks	Long-running pipelines
Logging	structlog	Structured pipeline logs
Reporting	Jinja2 + WeasyPrint	PDF report generation
Administration	pgAdmin	Database administration
Containerization	Docker Compose	Reproducible local deployment

4.3 Component Communication Workflow

Figure 4.2 traces how the five principal components — the React/Leaflet frontend, the FastAPI backend, the PostgreSQL/PostGIS/pgvector database, Google Earth Engine, and the Prithvi inference process — actually exchange data, distinguishing the calls that only ever happen offline (dashed, orange) from the calls a live user request makes (solid, blue).

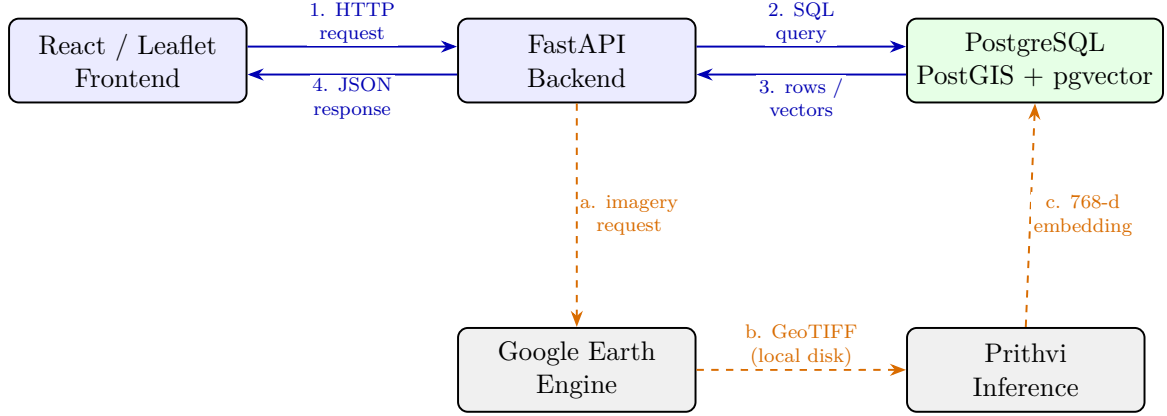


Figure 4.2: Component communication: solid blue is the online request path (frontend $\rightarrow$ backend $\rightarrow$ database, serving every request from precomputed rows); dashed orange is the offline pipeline path (backend $\rightarrow$ Earth Engine $\rightarrow$ Prithvi $\rightarrow$ database), triggered by Figure 4.1 as **BackgroundTasks**.

Neither path bypasses the backend: the frontend has no direct connection to the database, Earth Engine, or Prithvi, and the offline pipeline has no direct connection to the frontend. Every cross-component call passes through FastAPI, enforcing the offline/online separation in one place rather than by convention across every caller: the offline path (lake preparation, imagery acquisition, raster merging, spatial-cell extraction, embedding generation, temporal-feature preparation, vector indexing — the seven stages of Figure 3.1) is kept apart from the online path (lake lookup, map rendering, similarity query, analog profile retrieval, temporal-series retrieval, report generation), so Earth Engine and the foundation model never run during a user request.

4.4 PostgreSQL, PostGIS, and pgvector

PostgreSQL is the central persistent store. PostGIS keeps the original lake geometry, analysis-cell geometry, and spatial metadata in the database itself rather than converted into plain latitude/longitude tables, so intersection, containment, area, and map-query operations run directly as SQL. The pgvector extension stores the 768-dimensional Prithvi vectors alongside this spatial data, maintaining the association $\text{Embedding} \leftrightarrow \text{Subregion} \leftrightarrow \text{Lake} \leftrightarrow \text{Year}$ so that cosine-distance/similarity operations execute directly against the indexed database rather than exporting vectors to CSV or recomputing features.

4.5 Core Database Entities

The deployed schema, reached after nineteen Alembic migrations, consists of the tables in Table 4.2.

Table 4.2: Deployed database tables and current row counts.

Table	Rows	Role
<code>lakes</code>	11,130	HydroLAKES inventory with PostGIS geometry; <code>is_active</code> marks the 829-lake processing cohort
<code>lake_tiles</code>	19,046	Earth Engine export tiles per lake-year
<code>lake_images</code>	8,227	Merged lake-year mosaics with path and status
<code>sub_regions</code>	233,002	1 km analysis cells with geometry and coverage
<code>regions</code>	8,227	Pooled lake-year <code>vector(768)</code> embeddings
<code>lake_features</code>	8,227	Band statistics, indices, and texture per lake-year
<code>india_states</code>	36	State boundaries for spatial attribution
<code>reports</code>	—	Generated PDF report records

Despite its historical name (a legacy of the original design, preserved since renaming would invalidate existing embeddings for no functional gain), `regions` stores exactly one pooled 768-dimensional embedding per lake-year, enforced by a unique constraint on `(lake_id, year)` — the `vector(768)` column similarity search operates on. `lake_features`, the widest table in the schema, is the source of truth for measured reflectance: per-band statistics for B2–B7 and B8, pixel and coverage counts, cloud percentage, and GLCM texture, from which the forecasting layer reads only the band means and derives indices on demand. Keeping embeddings and measured reflectance in separate tables is what makes the decoder-free forecast possible: the embedding space answers *which lakes are analogs*, and the feature table independently answers *what measurably happened to them*, so the forecast never depends on inverting the foundation model.

4.6 Backend and Frontend

The FastAPI backend provides services for lake metadata lookup, nearest lake/grid-cell lookup, analog similarity search, temporal feature retrieval, forecast retrieval, report generation, and health/system status, separating cheap data retrieval from heavy batch processing so long operations are handled asynchronously.

The frontend is implemented using React, Vite, and TypeScript, with the map on Leaflet/React-Leaflet. A user selects or searches for a lake, views its geographic boundary and metadata, requests analogous ecosystems, visualizes their locations and similarity scores, and inspects their historical profiles alongside the forecast. Displayed lake boundaries and analog markers are drawn from the same PostGIS geometry used for the analysis itself, rather than a separately derived copy, which avoids the common failure mode where the analytical result and the displayed map disagree due to different coordinate transformations or numbering systems.

4.7 Asynchronous Processing and Progress Tracking

Imagery download, raster merging, feature extraction, and model inference all take far longer than an acceptable HTTP request, so each is launched through FastAPI's `BackgroundTasks` and returns immediately. A distributed task queue was considered and not adopted: the pipelines are operator-run batch jobs on a single machine, and the durable state a queue would provide already exists in the database, where every tile, image, cell, and feature row carries its own status. Progress is instead exposed through module-level state guarded by a `threading.Lock` (processed, skipped, and failed counts, elapsed time, throughput, projected completion), with each pipeline's block independent so concurrent runs report separately; the frontend polls the corresponding `/status` endpoint. Long jobs are also cancellable: a flag checked between work units means a cancelled run stops after the current lake-year rather than mid-write, and completed work stays committed.

4.8 Server-Side B8 Acquisition

Unlike B2–B7, which must be downloaded as pixels for Prithvi inference, B8 is needed only as summary statistics, so moving pixels at all would be wasted work. The B8 pipeline instead performs the reduction entirely on Earth Engine's servers using `Image.reduceRegions()`: lake polygons are batched fifteen at a time into a `FeatureCollection` and reduced against a single annual composite (reused across every lake in that year) with a combined mean/stdDev/median/minMax reducer, so only the resulting numbers are transferred and no imagery is ever downloaded. Transient failures are retried three times with exponential backoff, and a separate `b8_status` column tracks this pipeline independently so a B8 failure never marks the underlying feature row as failed — completed for all **8,227** lake-years in the cohort.

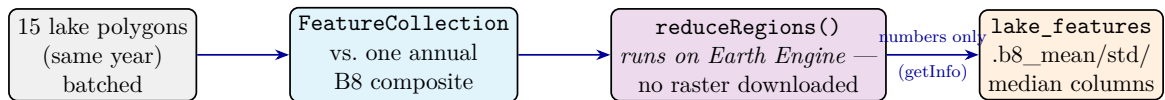


Figure 4.3: Stage 7 (Fetch B8): the only stage that never produces a local GeoTIFF — the zonal reduction runs on Earth Engine's servers against 15 batched polygons, and only reduced values cross the network into `lake_features`.

4.9 Automated PDF Report Generation

The application generates a per-lake PDF report through a Jinja2 template rendered to PDF by WeasyPrint. The report assembles lake metadata, a static map rendering of the lake and its analogs, historical index trajectories, the ranked analog table with similarity scores and matched windows, and the directional forecast table with its audit trail. This provides an offline deliverable for users who need a citable artifact rather than an interactive session.

4.10 Configuration Management

The processing pipeline reads experiment-specific values — Earth Engine project/authentication, date ranges, cloud thresholds, imagery buffer distance and tile size, analysis-cell size, coverage threshold, local storage paths, database credentials, and worker/batch settings — from environment configuration rather than hard-coding them throughout the processing code.

Chapter 5

Experimental Results and Engineering Evaluation

5.1 Deployed Processing Cohort

The HydroLAKES-derived inventory holds **11,130** Indian lakes. Of these, **829** are flagged `is_active` and constitute the fully processed cohort; the remainder are retained as inventory and map context but were not put through the expensive imagery and embedding stages. Selection favoured lakes large enough to yield meaningful multi-cell coverage at 10 m resolution, with a minimum area of 2 km².

Table 5.1: Deployed EcoTwin processing configuration.

Parameter	Deployed value
Lake source	HydroLAKES
Total inventory	11,130 lakes
Active processing cohort	829 lakes, 26 states
Cohort area range	2.00 – 843.22 km ² (mean 19.06)
Cohort elevation range	−6 – 4,985 m
Imagery window	2016–2025 (10 years)
Primary Sentinel collection	S2 SR Harmonized
Fallback collection	S2 Level-1C
Scene cloud-cover limit	20%
Composite method	Annual median
Imagery buffer	1,200 m
Imagery tile size	8,500 m
Analysis cell	1,000 m × 1,000 m
Coverage threshold	15%
Spatial resolution	10 m
Embedding model	Prithvi-EO-1.0-100M
Embedding dimension	768
Input bands	B2–B7 (6 bands)
Auxiliary band	B8 (statistics only)
Pooling	Coverage-weighted mean, L2-normalised
Similarity metric	Cosine over 5-year trajectories
Forecast horizon	3 years

5.2 Pipeline Completion Statistics

Table 5.2 reports the state of the production database. Every stage ran to completion with no residual failures.

Table 5.2: Final production statistics, queried directly from the deployed database.

Metric	Value	Status
Source lakes (HydroLAKES)	11,130	—
Active processing cohort	829	—
States covered	26	—
Lake-years attempted	8,290	—
Lake-years completed	8,227	99.2%
Imagery tiles generated	19,046	100% completed
Merged lake images	8,227	100% completed
Analysis cells generated	233,002	100% completed
Mean cells per lake-year	28.3	—
Embeddings indexed	8,227	100% non-null
Embedding failures	0	—
Feature records	8,227	100% completed
B8 statistics	8,227	100% completed
Mean lake coverage per image	35.6%	—
Mean valid pixels per lake-year	205,463	—

The **mean lake coverage of 35.6%** follows directly from the 1200m buffer and the design decision that an ecosystem representation should include shoreline and surrounding landscape, not water alone — a merged lake image is expected to be roughly one-third water. The **zero embedding failures** across 233,002 cell inferences reflects the pre-insertion validation gate (Section 3.10) combined with removal of an earlier stub-embedding fallback that had silently substituted a random vector on inference failure; that fallback was deleted, so inference failure now raises and no invalid vector reaches the database.

Early years have substantially less usable surface-reflectance coverage than recent ones (2016–2017 required special handling), which is why EcoTwin stores year-level processing status rather than assuming every lake has a complete ten-year sequence. The embedding pipeline (CPU batch size 8) is designed so the number of embeddings equals the number of valid analysis cells rather than the number of lakes, so a large lake generates several spatial embeddings and the vector index contains spatially localized representations rather than one vector per lake.

5.3 Similarity Search Performance

Table 5.3 reports measured search performance over the five Ramsar queries of Chapter 6, each executed against the full deployed index.

Table 5.3: Measured trajectory-similarity search performance.

Query lake	Candidates	Latency (ms)	Best raw cosine
Chilika	828	146	0.99691
Wular	828	141	0.99722
Loktak	828	148	0.99868
Tso Morari	828	160	0.99720
Vembanad	828	172	0.99830
Mean	828	153	0.99766

Each query evaluates roughly 4,900 candidate windows — 828 lakes with up to six windows each — at 768 dimensions, and completes in about 153 ms. This is comfortably interactive, and the index is fully scanned rather than approximated, so the returned ranking is exact.

5.4 The Score Compression Problem

Measured raw cosine scores across all five queries fall between **0.9969** and **0.9987**, with the tenth-ranked analog typically within **0.005** of the first — expected given that every vector shares one encoder and describes the same general scene type (Section 3.11), so *the absolute score is close to uninformative; only the ordering carries signal*. A similarity score therefore indicates closeness in the learned representation space, not identical ecological processes; ecological interpretation requires additional evidence from lake metadata, spatial maps, and temporal environmental indicators (Chapter 6). A meaningful absolute scale would require calibration against labelled ecosystem pairs (Section 8.2).

5.5 Temporal Profile Results

The temporal layer provides a second, independent axis of validation alongside spatial similarity: EcoTwin presents, for the query lake and its analogs, the full ten-year series of all five implemented indices (Table 2.3), observation availability, and the directional forecast with its contributing twins. Its value is demonstrated concretely by Wular Lake in Chapter 6, whose geographically unrelated top analogs nonetheless share its independently measured eutrophication trend.

5.6 Processing Reliability

The system was intentionally designed to tolerate partial failure. A lake-year can be pending, processing, completed, failed, or skipped because its output already exists,

which makes large multi-year extraction jobs restartable instead of requiring a complete restart after a single failed Earth Engine request.

5.7 Engineering Problems Encountered

Several recurring problems shaped the implementation. Large Earth Engine requests sometimes produced memory-limit errors, solved by reducing the size of each export region and moving part of the processing to local raster operations; parallel workers separately caused HTTP 429 and connection-pool saturation errors when too many requests were issued simultaneously, which pushed the pipeline toward smaller controlled worker counts, resumable jobs, and local reuse of already-generated imagery. The early Sentinel-2 Level-1C fallback exposed a different set of quality bands than other collections — requests that selected `MSK_CLASSI_CIRRUS`, `MSK_CLASSI_OPAQUE`, or `QA60` could fail depending on the actual image schema — which led to collection-aware preprocessing rather than a single hard-coded cloud-mask implementation. Merged rasters can also become much larger than individual source tiles, making compression, floating-point type control, tiled GeoTIFF writing, and local reuse important; at one point during development the local root filesystem reached 100% utilization, causing raster operations to fail with “No space left on device” and demonstrating that the pipeline must account for intermediate storage, not only final dataset size. Finally, Earth Engine authentication and access depend on external network services, and authentication failures or unreachable OAuth endpoints temporarily prevented pipeline execution — so the implementation separates authentication from per-lake processing and records failures instead of silently producing incomplete datasets.

Because the project was developed by multiple team members, a database consolidation procedure was also designed: each developer creates a PostgreSQL custom-format dump, which is restored into a temporary database, validated through a Python ETL process, and then merged into the production database. The procedure preserves PostGIS geometries, pgvector embeddings, indexes, constraints, and PostgreSQL data types, which is safer than CSV-based migration because CSV does not naturally preserve PostGIS geometry and vector types.

Chapter 6

Applied Analysis: Five Ramsar Wetlands

6.1 Purpose and Site Selection

This chapter runs the system end to end on five wetlands and reports what came back, including inconclusive results. Ramsar sites are ecologically significant and comparatively well-documented, so an independent account of site condition exists against which EcoTwin’s purely satellite-derived output can be sanity-checked. Five sites with complete ten-year records were selected, spanning a deliberately wide range of climate, altitude, and wetland type.

Table 6.1: The five Ramsar sites analysed. All five have complete 2016–2025 embedding and feature coverage.

Site	Lake ID	State	Area	Elev.	Type
Chilika	1506	Odisha	843.22	1	Brackish coastal lagoon
Tso Morari	1413	Ladakh	141.25	4522	High-altitude cold desert
Loktak	15510	Manipur	58.30	768	Freshwater, floating <i>phumdi</i>
Wular	14989	J&K	41.36	1574	Himalayan floodplain
Vembanad	15829	Kerala	10.14	0	Tropical backwater estuary

The spread — sea level to 4522 m, tropical estuary to Trans-Himalayan cold desert, two orders of magnitude in area — is deliberate: a system that only worked on lakes resembling one another would be exposed by it.

Every site was analysed under an identical protocol: a five-year query trajectory (2021–2025), a sliding-window search over all 828 other lakes, selection of the top five usable twins, delta extraction, fixed rank weights with per-year renormalization, and classification at $\tau = 0.005$ (Chapter 3), with anchor year 2025 and forecasts covering 2026–2028.

6.2 Chilika Lagoon, Odisha

Chilika is Asia’s largest brackish-water lagoon and the largest lake in the cohort (843 km²), separated from the Bay of Bengal by a barrier spit.

6.2.1 Retrieved Analogs

Table 6.2: Chilika — top five analogs (828 candidates, 146 ms).

#	Lake	State	Area	Raw cos.	Window
1	15704	Andhra Pradesh	12.28	0.99691	2017–2021
2	Govind Ballabh Pant	Uttar Pradesh	398.80	0.99646	2021–2025
3	177805	Gujarat	6.90	0.99554	2019–2023
4	15714	Telangana	27.56	0.99550	2018–2022
5	1509	Telangana	335.78	0.99536	2020–2024

Three of the five matched windows are *time-shifted*: the top analog matched on 2017–2021, so its 2022–2024 behaviour is already-observed evidence about a trajectory Chilika is only now traversing (Section 3.11).

6.2.2 Measured History

Table 6.3: Chilika — measured index history, 2016–2025.

Index	2016	2018	2020	2022	2024	2025
NDCI	0.0360	0.0323	0.0463	0.0518	0.0527	0.0415
NDVI-B7	−0.0450	−0.1469	−0.0865	−0.0686	−0.0491	−0.0806
NDWI	0.0911	0.2772	0.2169	0.1715	0.1578	0.1985
Turbidity	−0.0463	−0.1359	−0.1328	−0.1041	−0.1095	−0.1198
Red-edge slope	−2.3037	−2.9020	−2.5017	−2.7318	−2.2062	−2.3865

The persistently negative NDVI-B7 and positive NDWI are the expected signature of a large, genuinely open water body.

6.2.3 Forecast

Table 6.4: Chilika — directional forecast 2026–2028. Parenthesised values are weighted scores; n is contributing twins.

Index	2025	2026	2027	2028
NDCI	0.0415	→ (+0.0029, $n=5$)	→ (+0.0008, $n=4$)	↑ (+0.0391, $n=3$)
NDVI-B7	−0.0806	↑ (+0.0244, $n=5$)	↑ (+0.0406, $n=4$)	↑ (+0.0669, $n=3$)
NDWI	0.1985	↓ (−0.0205, $n=5$)	↓ (−0.0576, $n=4$)	↓ (−0.0950, $n=3$)
Turbidity	−0.1198	→ (−0.0032, $n=5$)	↑ (+0.0219, $n=4$)	↑ (+0.0333, $n=3$)
Red-edge	−2.3865	↑ (+0.2800, $n=5$)	↑ (+0.3180, $n=4$)	↑ (+0.2312, $n=3$)

6.2.4 Interpretation

The forecast is internally coherent: NDVI-B7 rises while NDWI falls by a comparable magnitude every year — expected, since both share B7, so vegetation encroaching on open water must raise one and lower the other. Rising red-edge slope and turbidity from 2027 reinforce the same reading: **gradual vegetative encroachment and reduction of open water**, with the declining twin count (5, 4, 3) signalling that 2028 should be read more cautiously than 2026.

6.3 Wular Lake, Jammu & Kashmir

Wular is one of Asia’s largest freshwater lakes, fed by the Jhelum in the Kashmir Valley, and among the most-documented degrading wetlands in India (siltation, shrinkage, weed proliferation).

6.3.1 Retrieved Analogs

Table 6.5: Wular — top five analogs (828 candidates, 141 ms).

#	Lake	State	Area	Raw cos.	Window
1	177198	Assam	2.95	0.99722	2021–2025
2	177700	West Bengal	2.64	0.99676	2021–2025
3	174785	Jammu & Kashmir	2.05	0.99653	2021–2025
4	176918	Assam	2.26	0.99595	2021–2025
5	177648	West Bengal	4.75	0.99582	2021–2025

The analogs are small floodplain water bodies in Assam and West Bengal — utterly unlike Wular in size, altitude, and climate, but sharing a *spectral trajectory* of shallow,

vegetation-choked water. The system matches ecological condition, not geography, as designed.

6.3.2 Measured History — The Clearest Signal in the Study

Table 6.6: Wular — measured index history, 2016–2025.

Index	2016	2018	2020	2022	2024	2025
NDCI	0.0372	0.1469	0.1866	0.2099	0.1769	0.2101
NDVI-B7	0.0621	0.1871	0.3419	0.3077	0.2662	0.2963
NDWI	−0.0198	−0.0927	−0.2224	−0.1943	−0.1766	−0.1878
Turbidity	−0.0424	−0.0961	−0.1294	−0.1206	−0.0940	−0.1149
Red-edge slope	0.8124	1.0597	4.1133	2.7214	2.6702	2.6200

This is the most pronounced trend recovered anywhere in the study. Over ten years NDCI rises **5.6-fold** ($0.037 \rightarrow 0.210$) and NDVI-B7 rises **4.8-fold** ($0.062 \rightarrow 0.296$), while NDWI falls to -0.188 and the red-edge slope more than triples — consistent across all ten annual composites, and unambiguous: **sustained conversion of open water to vegetated surface, with rising chlorophyll**. This is the closest thing to external validation the project obtained: EcoTwin, given no information beyond Sentinel-2 reflectance, independently recovered the degradation trajectory the lake is documented to be undergoing.

6.3.3 Forecast

Table 6.7: Wular — directional forecast 2026–2028.

Index	2025	2026	2027	2028
NDCI	0.2101	↑ (+0.0182, $n=5$)	↑ (+0.0147, $n=4$)	↑ (+0.0087, $n=3$)
NDVI-B7	0.2963	↑ (+0.0337, $n=5$)	↑ (+0.0168, $n=4$)	↓ (−0.0105, $n=3$)
NDWI	−0.1878	↓ (−0.0298, $n=5$)	↓ (−0.0119, $n=4$)	↓ (−0.0142, $n=3$)
Turbidity	−0.1149	↓ (−0.0063, $n=5$)	↓ (−0.0064, $n=4$)	↑ (+0.0261, $n=3$)
Red-edge	2.6200	↑ (+0.6184, $n=5$)	↓ (−0.0429, $n=4$)	↓ (−0.0244, $n=3$)

6.3.4 Interpretation

The forecast projects **continued eutrophication**: NDCI rises and NDWI falls in all three years, with NDCI increments shrinking year on year ($+0.018$, $+0.015$, $+0.009$) as an already-degraded system approaches saturation. The NDVI-B7 reversal in 2028 is worth flagging: it rests on three twins, its magnitude is barely twice the classification threshold, and it contradicts both the ten-year trend and 2028’s NDCI direction — the twin count is what lets a reader see this cell is weak.

6.4 Loktak Lake, Manipur

Loktak is the largest freshwater lake in northeast India, unique for its *phumdi* — floating mats of vegetation and organic matter supporting the world’s only floating national park — making its surface a natural stress test for an index set built on water/vegetation contrast.

6.4.1 Retrieved Analogs

Table 6.8: Loktak — top five analogs (828 candidates, 148 ms).

#	Lake	State	Area	Raw cos.	Window
1	177772	Madhya Pradesh	6.36	0.99868	2020–2024
2	178276	Maharashtra	9.00	0.99843	2020–2024
3	15539	Tripura	34.41	0.99840	2020–2024
4	15741	Karnataka	16.83	0.99822	2021–2025
5	178489	Andhra Pradesh	8.02	0.99804	2021–2025

Loktak produced the **highest raw similarity in the study** (0.99868); its third-ranked analog, in Tripura, is the only case where a top analog is both geographically near and climatically similar to the query.

6.4.2 Measured History

Table 6.9: Loktak — measured index history, 2016–2025.

Index	2016	2018	2020	2022	2024	2025
NDCI	0.1092	0.1040	0.1625	0.2171	0.1887	0.1590
NDVI-B7	0.1913	0.1819	0.3216	0.4075	0.3829	0.3242
NDWI	−0.0550	−0.0296	−0.1965	−0.2696	−0.2634	−0.1860
Turbidity	−0.1377	−0.1532	−0.1335	−0.1550	−0.1329	−0.1470
Red-edge slope	0.8915	0.9331	2.3458	3.3448	3.3716	2.6009

Loktak has the highest NDVI-B7 of any site studied, peaking at 0.4075 in 2022 (the *phumdi* mats), then partially receding through 2025 — a non-monotone pattern distinct from Wular’s sustained decline.

6.4.3 Forecast — and an Instructive Failure

Table 6.10: Loktak — directional forecast 2026–2028.

Index	2025	2026	2027	2028
NDCI	0.1590	↓ (−0.0086, $n=5$)	↑ (+0.0705, $n=2$)	? ($n=0$)
NDVI-B7	0.3242	→ (+0.0043, $n=5$)	↑ (+0.1694, $n=2$)	? ($n=0$)
NDWI	−0.1860	↑ (+0.0120, $n=5$)	↓ (−0.1893, $n=2$)	? ($n=0$)
Turbidity	−0.1470	↓ (−0.0161, $n=5$)	↑ (+0.0165, $n=2$)	? ($n=0$)
Red-edge	2.6009	↑ (+0.1294, $n=5$)	↑ (+0.1755, $n=2$)	? ($n=0$)

6.4.4 Interpretation

The 2028 column is entirely *uncertain* — correctly refusing to guess rather than a failed forecast. The cause is structural: three of Loktak’s five twins matched on windows ending in 2024, and the dataset ends in 2025, leaving $|\mathcal{A}_{k,3}| = 0$ for every index. Had *uncertain* been collapsed into *stable* — the earlier implementation’s behaviour — this column would have reported five confident “no change” predictions backed by no evidence at all (Section 3.13). The 2027 column carries a softer version of the same warning: two twins produce a large NDVI-B7 score (+0.1694), but the reported $n = 2$ signals that consensus is thin.

6.5 Tso Morari, Ladakh

Tso Morari is a high-altitude brackish lake at 4522 m in the Trans-Himalayan cold desert — the cohort’s environmental extreme, with minimal vegetation and glacial meltwater input.

6.5.1 Retrieved Analogs — The Strongest Validation

Table 6.11: Tso Morari — top five analogs (828 candidates, 160 ms).

#	Lake	State	Area	Raw cos.	Window
1	Pangong Tso	Ladakh	285.46	0.99720	2021–2025
2	178351	Maharashtra	8.52	0.98816	2021–2025
3	15661	Maharashtra	10.12	0.98781	2021–2025
4	178391	Maharashtra	3.06	0.98769	2021–2025
5	15828	Kerala	26.07	0.98599	2021–2025

This is the single most convincing retrieval result in the study. Out of 828 candidates spanning 26 states, the top-ranked analog for Tso Morari is **Pangong Tso** — the

other great high-altitude brackish lake of Ladakh, roughly 120 km away — recovered from reflectance alone, with no coordinates or lake-type metadata supplied. The score gap makes it stronger still: 0.99720 against rank 2’s 0.98816, a gap **five times** the spread between ranks 2 and 5 and sharply larger than the compression discussed in Chapter 5 — the embedding space is genuinely isolating high-altitude cold-desert lakes as a distinct class.

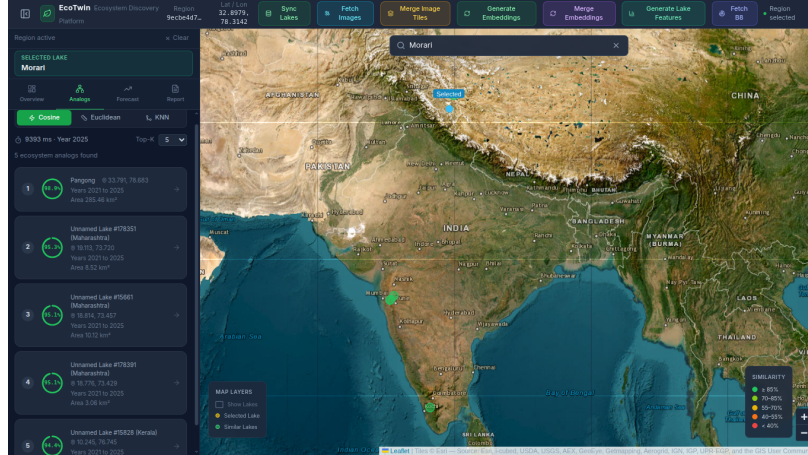


Figure 6.1: Live ranked-analog panel for Tso Morari, captured independently from the running application: Pangong returned as rank 1 at 98.9% displayed similarity, confirming Table 6.11.

6.5.2 Measured History

Table 6.12: Tso Morari — measured index history, 2016–2025.

Index	2016	2018	2020	2022	2024	2025
NDCI	−0.0396	−0.0212	−0.0417	−0.0450	−0.0337	−0.0376
NDVI-B7	−0.0907	−0.0649	−0.0805	−0.0653	−0.0339	−0.0509
NDWI	0.4804	0.2449	0.3921	0.3699	0.2700	0.3371
Turbidity	−0.4074	−0.1828	−0.3217	−0.3121	−0.2382	−0.2912
Red-edge slope	−0.0893	−0.2676	−0.1009	−0.0617	−0.0007	−0.0446

Tso Morari records the **highest NDWI** (0.48) and **most negative turbidity** (−0.41) in the study — the signature of exceptionally clear, vegetation-free, glacier-fed water, and the opposite extreme from Loktak.

6.5.3 Forecast

Table 6.13: Tso Morari — directional forecast 2026–2028.

Index	2025	2026	2027	2028
NDCI	−0.0376	↑ (+0.0204, $n=5$)	→ (+0.0046, $n=5$)	↓ (−0.0057, $n=5$)
NDVI-B7	−0.0509	↑ (+0.0360, $n=5$)	↑ (+0.0228, $n=5$)	↓ (−0.0177, $n=5$)
NDWI	0.3371	↓ (−0.0537, $n=5$)	↓ (−0.0216, $n=5$)	→ (−0.0035, $n=5$)
Turbidity	−0.2912	↑ (+0.0204, $n=5$)	↓ (−0.0058, $n=5$)	↑ (+0.0172, $n=5$)
Red-edge	−0.0446	↑ (+0.0499, $n=5$)	↑ (+0.2908, $n=5$)	↓ (−0.0522, $n=5$)

The live application’s Forecast panel (Figure 6.2) agrees with Table 6.13 on **direction for all fifteen cells**, despite drawing on 15 contributing twins rather than the 5 used offline — agreement despite a threefold difference in twin-pool size is evidence the classification is not an artifact of which handful of twins happens to be selected.

6.5.4 Interpretation

Tso Morari is the **only site where all five twins contribute to all three forecast years** ($n = 5$ throughout), the best-supported forecast in the study since its twins matched on early windows, leaving three full years of observed future. The projection is mild — modest NDVI-B7/NDCI increases, NDWI declining slightly then partially reversing — appropriately undramatic for an ecosystem with no significant vegetation dynamic, and corroborated by the live application run using an independently drawn, larger twin pool.

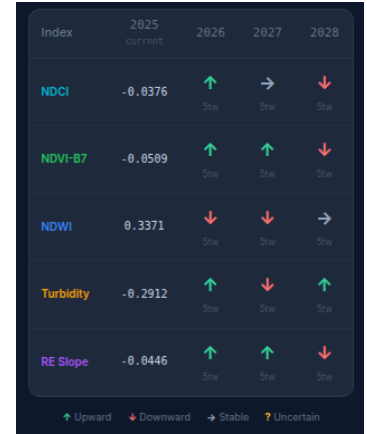


Figure 6.2: Live forecast summary for Tso Morari, 2026–2028, directly comparable to Table 6.13.

6.6 Vembanad, Kerala

Vembanad is the longest lake in India, the core of the Kerala backwater system — a tropical estuarine wetland at sea level, subject to tidal exchange and intense agricultural/urban pressure.

6.6.1 Retrieved Analogs

Table 6.14: Vembanad — top five analogs (828 candidates, 172 ms).

#	Lake	State	Area	Raw cos.	Window
1	176886	Assam	3.21	0.99830	2021–2025
2	15611	Maharashtra	11.75	0.99814	2020–2024
3	178503	Telangana	2.94	0.99791	2021–2025
4	15394	Uttarakhand	10.66	0.99779	2017–2021
5	178493	Telangana	2.64	0.99757	2020–2024

6.6.2 Measured History

Table 6.15: Vembanad — measured index history, 2016–2025.

Index	2016	2018	2020	2022	2024	2025
NDCI	0.0599	0.0454	0.0476	0.0546	0.0578	0.0500
NDVI-B7	0.0065	0.0302	0.0455	0.0417	0.0545	0.0365
NDWI	0.0900	0.0755	0.0696	0.0672	0.0559	0.0755
Turbidity	−0.0964	−0.1055	−0.1148	−0.1086	−0.1101	−0.1117
Red-edge slope	−0.7617	−0.2258	−0.0305	−0.1827	−0.0481	−0.1832

Vembanad is the most *stable* site in the study: every index varies within a narrow band across the decade (NDCI within ± 0.008 , NDWI within ± 0.02), consistent with a tidally flushed system buffered by continuous marine exchange.

6.6.3 Forecast — Degrading Confidence

Table 6.16: Vembanad — directional forecast 2026–2028.

Index	2025	2026	2027	2028
NDCI	0.0500	↓ (−0.0121, $n=5$)	↓ (−0.0488, $n=3$)	? ($n=1$)
NDVI-B7	0.0365	↓ (−0.0132, $n=5$)	↓ (−0.0994, $n=3$)	? ($n=1$)
NDWI	0.0755	→ (−0.0024, $n=5$)	↑ (+0.1119, $n=3$)	? ($n=1$)
Turbidity	−0.1117	↑ (+0.0175, $n=5$)	↓ (−0.0085, $n=3$)	? ($n=1$)
Red-edge	−0.1832	↓ (−0.0947, $n=5$)	↓ (−1.2968, $n=3$)	? ($n=1$)

6.6.4 Interpretation

Vembanad shows the confidence-degradation pattern in its clearest form: $n = 5, 3, 1$ across the three years. The 2028 column, with exactly one contributing twin, is correctly withheld as *uncertain* — a single lake’s behaviour is not a consensus. The 2027 red-edge score (-1.2968) is dominated by one twin with an extreme delta: its direction is defensible but the magnitude should not be read as calibrated. The 2026 column, resting on all five twins, is the reliable row: mild declines in NDCI and NDVI-B7 with NDWI flat, consistent with the decade of measured stability.

6.7 Cross-Site Synthesis

Table 6.17: Cross-site summary. Confidence is the number of contributing twins across the three horizon years.

Site	Best raw	Latency	Dominant measured signal	Conf.
Chilika	0.99691	146 ms	Mild vegetative encroachment	5,4,3
Wular	0.99722	141 ms	Strong eutrophication ($5.6\times$ NDCI)	5,4,3
Loktak	0.99868	148 ms	<i>Phumdi</i> peak then partial recession	5,2,0
Tso Morari	0.99720	160 ms	Stable clear cold-desert water	5,5,5
Vembanad	0.99830	172 ms	Decadal stability (tidally buffered)	5,3,1

Five findings emerge from analysing the sites together. **The embedding space encodes real ecological structure:** the Tso Morari–Pangong pairing, recovered from reflectance alone with a score gap five times the typical inter-rank spread, is not explicable as coincidence. **Condition dominates geography in analog retrieval:** Wular’s analogs are small Assamese and Bengali floodplain lakes, matching ecological state as intended, though it means the map view alone can mislead without the index history. **The index set discriminates across the full range,** from Tso Morari’s NDWI of 0.48 with essentially no vegetation to Loktak’s NDVI-B7 of 0.41 under floating mats — the band-honest substitution of B7 for B8 did not cost discriminative power. **Forecast confidence is bounded by data recency, not model quality:** every degraded forecast — Loktak’s empty 2028, Vembanad’s single-twin 2028 — traces to twins whose matched windows end near 2025, a dataset-extent limitation that self-corrects as further years are ingested. Finally, **the system declines to answer when it should:** across 75 forecast cells ($5 \text{ sites} \times 5 \text{ indices} \times 3 \text{ years}$), the outcomes divide as shown in Table 6.18.

Table 6.18: Distribution of the 75 forecast cells across the five Ramsar sites.

Classification	Cells	Share
Up ($\uparrow$)	31	41.3%
Down ($\downarrow$)	27	36.0%
Stable ($\rightarrow$)	7	9.3%
Uncertain (?)	10	13.3%
Total	75	100%

Thirteen percent of cells returned *uncertain* rather than a direction: a method that always answers cannot be distinguished from one that guesses, so these ten abstentions are evidence the confidence machinery is load-bearing. The near-balance of up and down classifications is also reassuring against directional bias.

Chapter 7

Discussion

7.1 Strengths and Limitations

The system’s central strengths, established in earlier chapters, are scalable offline preprocessing executed to completion with zero residual failures (Chapter 5); trajectory-based retrieval that matches ecological *condition* rather than single-year appearance, demonstrated by the Tso Moriri–Pangong pairing (Chapter 6); and a decoder-free, fully auditable forecasting layer that reports *uncertain* rather than guessing when evidence is thin (a 13.3% abstention rate across 75 forecast cells).

Its limitations are correspondingly structural. The ten-year window is not uniformly complete (Table 2.2), and Sentinel-2’s optical sensing means cloud cover reduces valid observations (mean 64.4% before masking). A high embedding similarity does not guarantee ecological equivalence — only the *ranking* is meaningful, not the compressed absolute score (Chapter 5) — and every result is conditional on the fixed configuration of Table 5.1, not a configuration-independent measurement. The clearest limitation is that a twin whose matched window ends near 2025 has little observed future to transfer, which emptied Loktak’s $h = 3$ forecast and thinned Vembanad’s (Chapter 6) — not a flaw in the classification rule, which correctly reported *uncertain*, but a consequence of dataset extent that self-corrects as further years are ingested. Finally, the Ramsar case studies were checked against publicly documented site condition rather than a quantitative field survey, which is corroborating evidence, not calibrated validation; formal hindcasting (Section 3.13.2) is the next validation step precisely because it does not require such external data.

Chapter 8

Conclusion and Future Work

8.1 Conclusion

EcoTwin evolved from a conceptual ecosystem-comparison idea into a complete, fully executed geospatial retrieval and forecasting system: HydroLAKES and Sentinel-2 feed a buffered, tiled acquisition pipeline into Prithvi embeddings, coverage-pooled to lake level and indexed in pgvector, from which a five-year sliding-window trajectory search ranks analogs and a decoder-free forecasting layer transfers their measured index changes onto the target with a full audit trail — executed to completion over 829 lakes, ten years, and zero outstanding failures (Chapter 5).

Chapter 6 demonstrated the complete pipeline on five Ramsar wetlands, returning a striking unsupervised validation (Tso Morari’s top analog is Pangong Tso, the only other high-altitude Ladakhi lake in the cohort), an independently corroborated degradation signal at Wular, and, equally important, a forecasting method that correctly declined to predict in 10 of 75 cells where evidence was insufficient rather than manufacturing a confident answer. The project’s most significant contribution is therefore the integration of geospatial data engineering, foundation-model inference, vector retrieval, temporal analysis, and visualization into one reproducible workflow carried through to applied results, demonstrating along the way — through the abandoned embedding-decoder design (Section 3.10) — that transferring measured evidence is often more defensible than reconstructing it from an uncalibrated inversion.

8.2 Future Work

Priority items are executing the hindcasting protocol (Section 3.13.2) once further years extend the dataset past its current single-cutoff limitation; extending ingestion beyond 2025 so recently-matched twins accumulate the future years needed to resolve *uncertain* cells; adding independent field-ecological validation for the Ramsar sites; and migrating trajectory search to pgvector’s native ANN operators, since the current brute-force scan will not remain interactive as the cohort grows. Further directions include calibrating the cosine-similarity display scale against labelled ecosystem pairs, attention- or learned-weighted pooling in place of coverage-weighted mean, growing the active cohort beyond 829 lakes, reducing monsoon-driven cloud gaps in temporal

compositing, and deploying shared production infrastructure for multi-user access.

8.3 Final Remarks

EcoTwin’s central idea is to use historically observed ecosystems as computationally retrievable analogs rather than treating every ecosystem as an isolated forecasting problem. The Ramsar case studies show that this idea, carried through a fully processed 829-lake, ten-year cohort, produces results that are specific, checkable against independent knowledge of the sites, and — where the evidence ran out — honest about the limits of what it can currently claim.

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